The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Observer-Relevant Pullback Body-Response Core
Necessity / Uniqueness Theorem

Abstract

The current active-primary theorem already removed the full observer-relevant pullback body-operator core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$ as the primitive stopping point of the active section-centered hidden-response line. Benchmark-facingly, the separate ambient operator-family presentation is now recovered from the smaller observer-relevant pullback body-response core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$. The current benchmark criterion therefore made the honest next move explicit: either derive or justify that body-response core from still more primitive explicit substrate structure, or, if no such clean deeper law is yet in hand, prove a sharper theorem strictly inside that smaller core.

The present note takes that honest internal route. It proves that on the current section-centered hidden-response route the observer-relevant pullback body-response core is already necessary and unique at benchmark scope up to body-response-faithful equivalence. More precisely, any benchmark-faithful reduction that preserves the fixed proto-section benchmark base, the current body-response recovery route, and the benchmark one-metric observer package must determine the same section-centered displacement-body datum together with the same body-restricted pullback response. Conversely, nothing beyond that core is needed at current benchmark scope, because the earlier pullback-body-response-core collapse theorem already shows that all present downstream benchmark-facing uses factor through that smaller core.

This is not yet a completed first-principles derivation of GR from substrate microphysics. Its positive result is narrower and more honest. It removes the remaining presentational freedom at the current body-response frontier on the recorded route. The next primary manuscript must therefore derive or uniquely force the current body-response core itself, or some nontrivial constituent or relation inside it, from still more primitive explicit substrate structure. Another same-output deeper relay or a re-expansion back to the larger pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations is no longer the right front-line move.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity now has a sharper task than merely going deeper again. The current pullback-body-response-core collapse theorem already spent the larger observer-relevant pullback body-operator core as the primitive stopping point on the active section-centered hidden-response route. Once the actual displacement body together with the quadratic pullback response realized on that body are fixed, the full ambient operator-family presentation is recovered and all current downstream benchmark-facing consequences follow unchanged. The present frontier is therefore already internal to the body-response core itself.

That status leaves only two honest options. One can derive or justify the current body-response core from still more primitive explicit substrate structure in a way that genuinely changes benchmark-facing status. Or, if that deeper forcing law is not yet cleanly available, one can prove a sharper internal theorem that removes remaining presentational freedom strictly inside the current body-response core. The present note takes the second route. Its claim is that, on the currently recorded route, the pullback body-response core is already the unique minimal benchmark-facing datum up to body-response-faithful equivalence.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen the larger pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations as if they were again independently live. Its narrower benchmark-facing role is to prove that the current body-response core is already the unique minimal datum of the recorded route at current scope.

2. Fixed Inputs from the Current Body-Response Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded proto-section benchmark base and current route)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base* $\mathfrak{Q}^{\mathrm{sub,proto,secgen}}$*, in particular the recorded section body* $\mathcal{B}_\sigma^{\mathrm{psec}}$*;*

2. *the current observer-relevant pullback body-response core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$*, the current observer-relevant pullback body-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}$*, the current observer-relevant pullback-operator core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}$*, the current observer-relevant pullback-quadratic core* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}$*, the current section-centered residual-normal-form realization* $\mathfrak{R}^{\mathrm{sub,proto,presec,res}}$*, the current observer-relevant pre-proto-section minimizer-section core* $\mathfrak{R}^{\mathrm{sub,proto,presec,sec}}$*, and the current benchmark-faithful pre-proto-section package* $\mathfrak{R}^{\mathrm{sub,proto,presec}}$*;*

3. *the previously recorded fact that, once a benchmark-faithful observer-relevant pullback body-response core over the fixed proto-section benchmark base is available, the current body-operator-core realization, the current pullback-operator-core realization, the current pullback-quadratic-core realization, the current section-centered residual-normal-form realization, the current pre-proto-section package, the current minimizer-section core, and the previously recorded seed, root, foundation, ground, origin, precursor, lift, shell-restriction, split-core, selector-law, combined-response, ambient-package, trace-core, exact-shell-affine-nucleus, continuity-Hessian compressed core, and benchmark one-metric observer-package consequences follow unchanged;*

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Observer-relevant pullback body-response core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful observer-relevant pullback body-response core $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$ *for sector* σ *consists of:*

1. *a compact convex section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{pppresec}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{pppresec}}$$

   whose fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y) \equiv \{v \in V_\sigma^{\mathrm{pppresec}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}\}$$

   is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

2. *the body-restricted pullback response*

$$\mathscr{R}_\sigma^{\mathrm{pppresec,pb,body}} : \mathcal{D}_\sigma^{\mathrm{pppresec}} \to [0, \infty),$$

   smooth in y*, fiberwise quadratic in* v*, and satisfying*

$$\mathscr{R}_\sigma^{\mathrm{pppresec,pb,body}}(y, v) = 0 \qquad \Longleftrightarrow \qquad v = 0$$

   on every fiber;

3. *benchmark faithfulness in the sense of Definition 2(4).*

From this core define the recovered ambient displacement space by

$$\widehat{V}_{\sigma}^{\mathrm{pppresec,pb,bodyresp}} \equiv \operatorname{span}\left(\bigcup_{y\in\mathcal{B}_{\sigma}^{\mathrm{psec}}} \mathcal{D}_{\sigma}^{\mathrm{pppresec}}(y)\right) \subset V_{\sigma}^{\mathrm{pppresec}}.$$

Remark 1. *Definition 3 is the current minimal data package identified by the preceding collapse theorem. The present note asks whether, on the recorded route itself, any smaller genuinely benchmark-facing datum still remains below it.*

Proposition 1 (What the preceding collapse theorem already proves)**.** *At current scope, the earlier pullback-body-response-core collapse theorem already proves all of the following.*

1. *The ambient displacement space is recovered canonically from the fiberwise span of the displacement body.*
2. *The body-restricted pullback response determines the unique compatible positive self-adjoint pullback operator family on that recovered space.*
3. *The full pullback body-operator core is reconstructed from the pullback body-response core together with the fixed proto-section benchmark base.*
4. *All current benchmark-facing downstream uses of the body-response frontier factor through* $\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}$.

Proof. This is exactly the content of the earlier theorem collapsing the observer-relevant pullback body-operator core to the observer-relevant pullback body-response core and of its downstream factorization statement. □

3. Body-Response-Faithful Benchmark Reductions

Definition 4 (Body-response-faithful benchmark reduction)**.** *A* body-response-faithful benchmark reduction *of the current section-centered hidden-response route is a reduction* $\mathfrak{B}_{\mathrm{bodyresp}}$ *from deeper explicit substrate variables to the observer-facing sector such that, after admissible elimination of auxiliary or substrate-only variables and for each* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*,*

1. *it preserves the fixed proto-section benchmark base of Definition 2(1);*
2. *it determines a benchmark-faithful candidate body-response datum*

   $$\mathfrak{D}_{\mathrm{bodyresp}}(\mathfrak{B}_{\mathrm{bodyresp}}, \sigma) = (\widetilde{\mathcal{D}}_{\sigma}, \widetilde{\mathscr{R}}_{\sigma})$$

   of the same form as Definition 3;
3. *it preserves the current route in the precise benchmark-facing sense that the candidate datum reconstructs a benchmark-faithful pullback body-operator core and hence the same current pullback-operator core, pullback-quadratic core, section-centered residual-normal-form realization, pre-proto-section package, continuity-Hessian compressed core, and benchmark one-metric observer package;*
4. *it introduces no second operative metric, no enlarged low-energy matter law, and no change to the ordinary matter route.*

Remark 2. *Definition 4 is intentionally route specific. The theorem of this note concerns the active recorded section-centered hidden-response route, not every imaginable future redesign that changes the route itself.*

Definition 5 (Body-response-faithful equivalence)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. Two candidate body-response data*

$$(\mathcal{D}_\sigma^{(1)}, \mathscr{R}_\sigma^{(1)}), \qquad (\mathcal{D}_\sigma^{(2)}, \mathscr{R}_\sigma^{(2)})$$

over the fixed proto-section benchmark base are body-response-faithfully equivalent *if, for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, there is a linear isomorphism*

$$\Phi_\sigma(y) : \mathrm{span}(\mathcal{D}_\sigma^{(1)}(y)) \xrightarrow{\cong} \mathrm{span}(\mathcal{D}_\sigma^{(2)}(y))$$

such that

$$\Phi_\sigma(y)(\mathcal{D}_\sigma^{(1)}(y)) = \mathcal{D}_\sigma^{(2)}(y)$$

and

$$\mathscr{R}_\sigma^{(2)}(y, \Phi_\sigma(y)v) = \mathscr{R}_\sigma^{(1)}(y, v) \qquad \text{for every } v \in \mathcal{D}_\sigma^{(1)}(y).$$

Definition 6 (Minimal observer-relevant body-response datum)**.** *For a body-response-faithful benchmark reduction* $\mathfrak{B}_{\mathrm{bodyresp}}$*, a datum* $\mathfrak{D}_{\mathrm{bodyresp}}(\mathfrak{B}_{\mathrm{bodyresp}})$ *is a* minimal observer-relevant body-response datum *if:*

1. *it suffices to determine every benchmark-facing derivation statement available on the current section-centered hidden-response route;*
2. *no proper subdatum of it still suffices at that same benchmark-facing scope.*

4. Necessity of the Current Body-Response Constituents

Proposition 2 (The displacement-body datum is necessary)**.** *Every minimal observer-relevant body-response datum for a body-response-faithful benchmark reduction must determine the section-centered displacement-body datum on the current route up to body-response-faithful equivalence.*

Proof. The current route does not use an abstract quadratic response with unspecified domain. It uses the recorded section-centered displacement body on which the response is actually realized. By Definition 3, that body records the admissible displacement fibers, contains the zero section, and has nonempty interior in each fiber. Proposition 1(1) then uses that body to recover the ambient displacement space itself. If the displacement-body datum were omitted, neither the domain of the current body response nor the recovered ambient-space step of the current route would be fixed. Therefore no minimal benchmark-facing datum on this route may omit the displacement-body constituent. □

Proposition 3 (The body-restricted pullback response is necessary)**.** *Every minimal observer-relevant body-response datum for a body-response-faithful benchmark reduction must determine the body-restricted pullback response on the current route up to body-response-faithful equivalence.*

Proof. The current route does not recover the pullback body-operator core, the pullback-operator core, or the remaining downstream benchmark-facing package from the displacement body alone. Proposition 1(2) uses the body-restricted quadratic response to recover the unique compatible

pullback operator family, and Proposition 1(4) then carries all current downstream benchmark-facing uses through that recovery. If the response constituent were omitted, the present route would no longer determine the operator recovery or the downstream package. Therefore no minimal benchmark-facing datum on this route may omit the response constituent. □

Proposition 4 (The current body-response core is sufficient)**.** *The current observer-relevant pullback body-response core* $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$ *is sufficient to determine every benchmark-facing derivation statement available on the current section-centered hidden-response route.*

Proof. By Proposition 1(4), all current benchmark-facing downstream uses of the present frontier already factor through $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$. Therefore the current body-response core is sufficient at benchmark scope on the recorded route. □

5. Main Theorem

Theorem 1 (Observer-relevant pullback body-response core necessity / uniqueness on the current route)**.** *For every body-response-faithful benchmark reduction* $\mathfrak{B}_{\text{bodyresp}}$ *of the current section-centered hidden-response route, the observer-relevant pullback body-response core* $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$ *of Definition 3 is necessary and unique at benchmark scope up to body-response-faithful equivalence. More precisely:*

1. *every minimal observer-relevant body-response datum* $\mathfrak{D}_{\text{bodyresp}}(\mathfrak{B}_{\text{bodyresp}})$ *must determine both the displacement-body datum and the body-restricted pullback response on the current route;*
2. $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$ *itself is sufficient to determine every benchmark-facing derivation statement available on that route;*
3. *any two minimal observer-relevant body-response data are body-response-faithfully equivalent to* $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$*, and hence to each other.*

Proof. Item (1) is the combined content of Propositions 2 and 3. Those propositions show that no minimal observer-relevant datum on the current route may omit either the displacement-body constituent or the body-restricted pullback-response constituent.

Item (2) is Proposition 4. The earlier collapse theorem already proves that all current benchmark-facing downstream uses factor through $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$.

For item (3), let $\mathfrak{D}^{(1)}_{\text{bodyresp}}$ and $\mathfrak{D}^{(2)}_{\text{bodyresp}}$ be two minimal observer-relevant body-response data for body-response-faithful benchmark reductions of the current route. By item (1), each must determine the same kind of displacement-body and body-response constituents through which the route recovers the benchmark-facing downstream package. By item (2), neither needs anything beyond those constituents at current benchmark scope. Therefore each is body-response-faithfully equivalent to $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$ in the sense of Definition 5. Hence they are body-response-faithfully equivalent to each other as well. □

Corollary 1 (No smaller benchmark-facing datum remains on the current route)**.** *On the current section-centered hidden-response route, there is no genuinely smaller benchmark-facing datum below* $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$ *up to body-response-faithful equivalence.*

Proof. If a smaller datum existed, it would either omit the displacement-body constituent or the body-restricted pullback-response constituent proved necessary by Propositions 2 and 3, or else merely rename the same constituents and therefore remain body-response-faithfully equivalent to $\mathfrak{S}^{\text{sub,proto,pppresec,pb,bodyresp}}$. The first option contradicts necessity, and the second is not genuinely smaller at benchmark scope. □

Remark 3. *Corollary 1 is the benchmark-facing sense in which the current body-response frontier is now exhausted internally on the recorded route. The route no longer has live benchmark-facing freedom in another body-response-core restatement or in a re-expansion to a larger presentation.*

6. What Must Be Derived Next

Corollary 2 (Primary post-uniqueness burden)**.** *After Theorem 1, the next primary manuscript below the current frontier must do at least one of the following:*

1. *derive the current displacement-body datum from still more primitive explicit substrate structure rather than preserving it as already-recorded route data;*
2. *derive the current body-restricted pullback response from still more primitive explicit substrate structure rather than preserving it as already-recorded route data;*
3. *prove a sharper necessity, uniqueness, or compatibility theorem for some nontrivial constituent or relation inside the present body-response core;*
4. *do one of the previous three while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.*

Another same-output deeper relay that merely preserves the same body-response core, or any re-expansion back to the larger pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations, is not primary by default.

Proof. By Theorem 1 and Corollary 1, the current body-response core is already the unique minimal benchmark-facing datum on the recorded route up to body-response-faithful equivalence. Therefore a new primary manuscript must derive or sharpen some constituent or relation inside that datum. If it does not, then it preserves the same benchmark-facing core and is neutral at project-status level. □

Proposition 5 (The benchmark package remains fixed)**.** *The present necessity / uniqueness theorem preserves the benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definitions 2(4), 3(3), and 4(4). The present note removes only the remaining presentational freedom at current body-response scope on the recorded route. □

7. Conclusion

The positive result of this note is narrower than a completed substrate derivation of GR but sharper than another routing reminder. On the current section-centered hidden-response route,

the observer-relevant pullback body-response core is now proved to be necessary and unique at benchmark scope up to body-response-faithful equivalence.

That core consists of the section-centered displacement-body datum together with the body-restricted pullback response realized on that body, under the same benchmark-faithfulness boundary. By the theorem of this note, no smaller genuinely benchmark-facing datum remains below it on the recorded route. The current pullback-body-response-core collapse theorem remains preserved mainline history because it first moved the live burden below the larger body-operator-core presentation. The present necessity / uniqueness theorem then spends the remaining presentational freedom at that smaller frontier.

This sharpens the honest next burden. A new primary manuscript must no longer merely show that an even deeper substrate layer reproduces the same body-response core or re-expands to a larger already-spent presentation. It must derive or uniquely force the current body-response core itself, or some nontrivial constituent or relation inside it, from still more primitive explicit substrate structure while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.